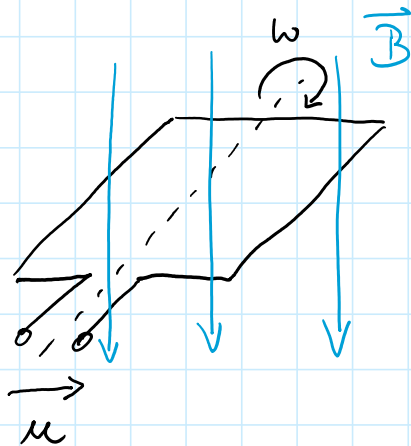
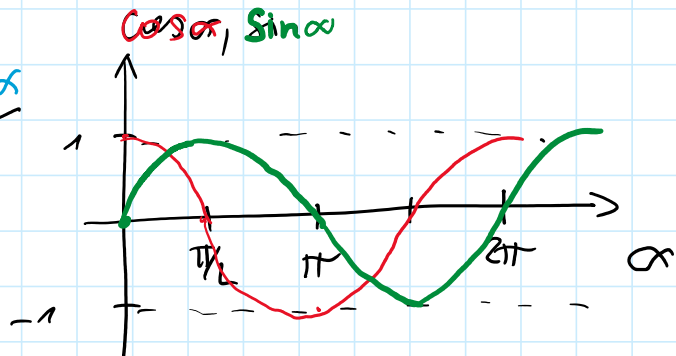
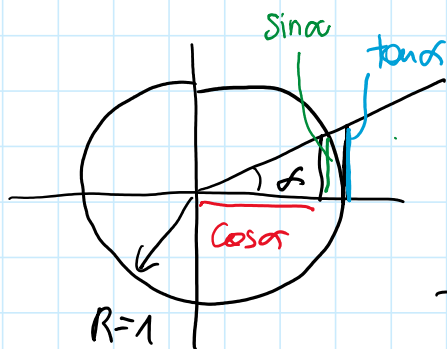




$$\omega = 2\pi f$$

$$u = \hat{u} \cos(\omega t + \frac{\pi}{2})$$



WENNZAHLEN

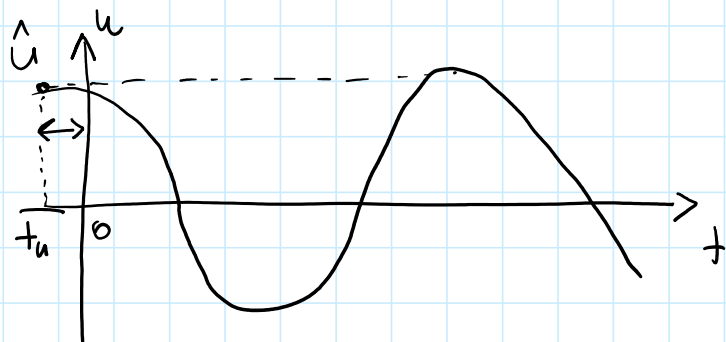
$$u(t) = \hat{u} \cos(\omega t + \varphi_u)$$

$\hat{u}$... Amplitude

$$u_{pp} = 2 \hat{u}$$

φ_u ... Nullphasenwinkel

$$\varphi_u > 0 \quad \varphi_u < 0$$



$$t_u = -\frac{\varphi_u}{\omega} \dots \text{Nullphasenzeit}$$

PHASENLAGE

$$\varphi_u = 0 \rightarrow \text{nullphasig}$$

$$\varphi_1 > \varphi_2 \rightarrow u_1 \text{ eilt } u_2 \text{ voraus}$$

$$\varphi_{12} = \varphi_1 - \varphi_2 \rightarrow \text{Phasenverschiebung}$$

MITTELWERTE

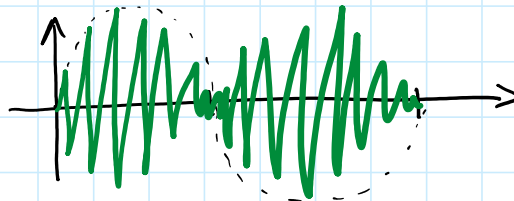
Mittelwerte

- Gleichwert $\bar{u} = \frac{1}{T} \int_0^T u \, dt = 0$
- Effektivwert $U = \sqrt{\frac{1}{T} \int_0^T u^2 \, dt} = \frac{\hat{u}}{\sqrt{2}}$
- Gleichrichtwert $|\bar{u}| = \frac{1}{T} \int_0^T |u| \, dt = \frac{2}{\pi} \hat{u}$

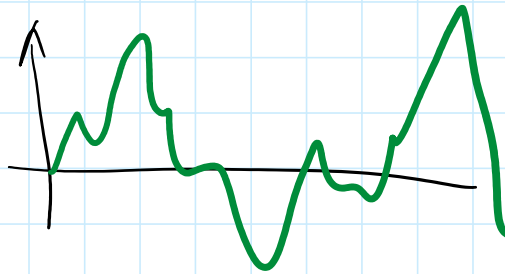
Überlagerungen

- $\omega_1 = \omega_2 \rightarrow u_3 = u_1 + u_2$ sinusförmig
- $\omega_1 \approx \omega_2 \rightarrow \omega = \frac{\omega_1 + \omega_2}{2} \quad \Delta\omega = \frac{\omega_1 - \omega_2}{2}$

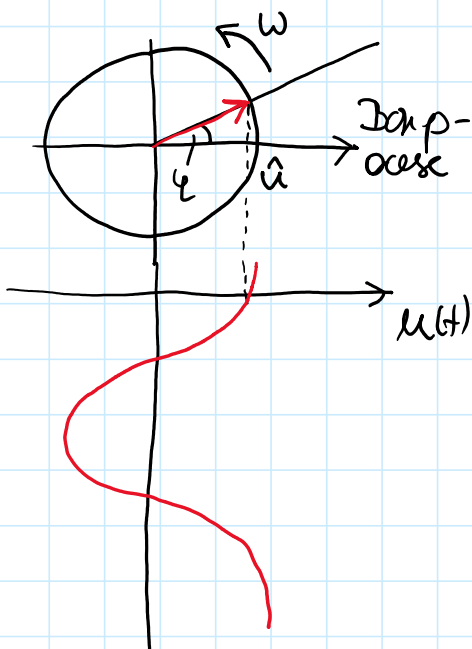
$u_1 = u_2$



- $\omega_1 \neq \omega_2 \rightarrow u_3$ keine sinusförmige Größe

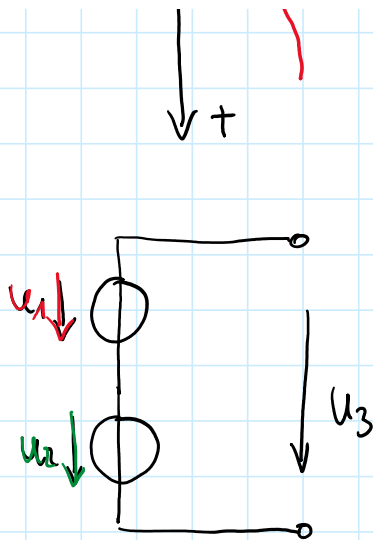


Zeit-Darstellung



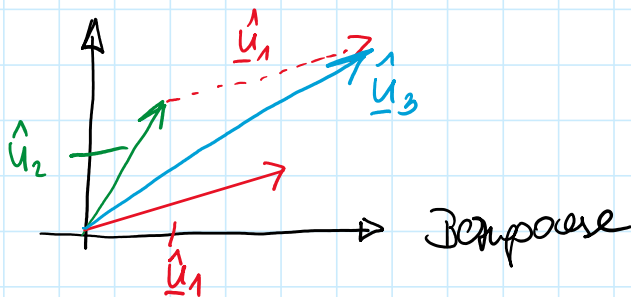
$$\varphi(t) = \varphi + \omega t$$

$$u = \hat{u} \cdot \cos(\omega t + \varphi)$$

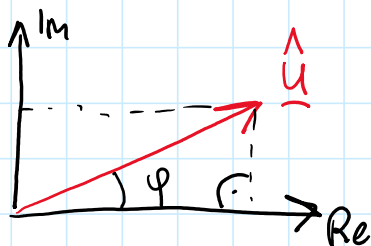


u_1	u_2
ω	ω
$\hat{u}_1$	$\hat{u}_2$
φ_1	φ_2

$$u_3 = \hat{u}_1 \cos(\omega t + \varphi_1) + \hat{u}_2 \cos(\omega t + \varphi_2)$$



KOMPLEXE SYMBOLE



$$j = \sqrt{-1}$$

$$\operatorname{Re}\{\underline{\hat{u}}\} = \hat{u} \cdot \cos(\varphi)$$

$$\operatorname{Im}\{\underline{\hat{u}}\} = \hat{u} \cdot \sin(\varphi)$$

$$\varphi = \arctan \frac{\operatorname{Im}\{\underline{\hat{u}}\}}{\operatorname{Re}\{\underline{\hat{u}}\}}$$

$$\hat{u} = \sqrt{\operatorname{Re}\{\underline{\hat{u}}\}^2 + \operatorname{Im}\{\underline{\hat{u}}\}^2}$$

$$u(t) = \hat{u} \cos(\omega t + \varphi) + j \hat{u} \sin(\omega t + \varphi)$$

$$\downarrow e^{j\varphi} = \cos \varphi + j \sin \varphi \quad \text{"Eulersche Formel"}$$

$$= \hat{u} e^{j(\omega t + \varphi)} \quad \dots \text{vollstndiges komplexes Symbol}$$

$$\underline{\hat{u}} = \hat{u} e^{j\varphi} = \hat{u} \angle \varphi$$

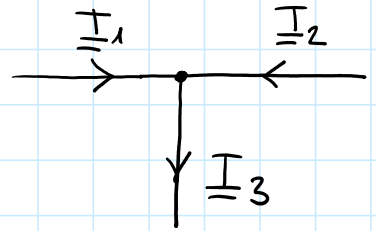
$$\underline{U} = U e^{j\varphi} = U \angle \varphi \quad (\text{sprich } U \text{ versor } \varphi)$$

$$\omega = 2\pi f = \sum_{n=1}^{\infty} \frac{1}{n} \quad \text{r.}$$

$$e = 2,71... = \sum_{k=0}^{\infty} \frac{1}{k!} \quad \mathbb{C}$$

$$\underline{I}_1 = 1,5 \text{ A} \angle 0^\circ \rightarrow i_1 = \sqrt{2} \cdot 1,5 e^{j(\omega t + 0^\circ)} \text{ A}$$

$$\underline{I}_2 = 0,5 \text{ A} \angle 120^\circ \rightarrow i_2 = \sqrt{2} \cdot 0,5 e^{j(\omega t + 120^\circ)}$$



$$\underline{I}_1 = 1,5 \text{ A} + j 0 \text{ A}$$

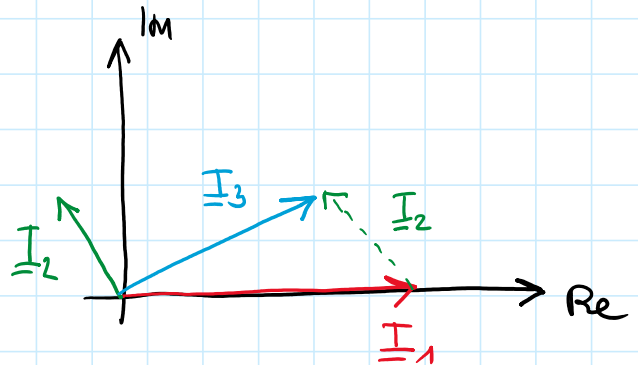
$$\underline{I}_2 = -0,25 \text{ A} + j 0,433 \text{ A}$$

$$\underline{I}_3 = \underline{I}_1 + \underline{I}_2 =$$

$$= 1,5 \text{ A} - 0,25 \text{ A} + j 0,433 \text{ A} =$$

$$= 1,25 \text{ A} + j 0,433 \text{ A} =$$

$$= \underline{1,32 \text{ A} \angle 19,1^\circ}$$

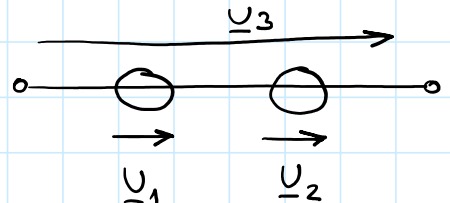


$$f = 50 \text{ Hz}$$

$$\underline{U}_3 = ?$$

$$\underline{U}_1 = 4,7 \text{ V} \angle 30^\circ$$

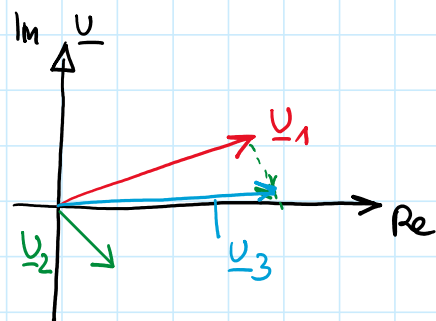
$$\underline{U}_2 = 2 \text{ V} \angle -45^\circ$$



$$\underline{U}_3 = \underline{U}_1 + \underline{U}_2 = 4,7 \text{ V} \angle 30^\circ + 2 \text{ V} \angle -45^\circ =$$

$$= 4,07 \text{ V} + j 2,35 \text{ V} + 1,414 \text{ V} - j 1,414 \text{ V} =$$

$$= 5,484 \text{ V} + j 0,936 \text{ V} = \underline{5,563 \text{ V} \angle 9,68^\circ}$$



$$u_1 = \sqrt{2} \cdot 4,7 \text{ V} \cdot \cos(2\pi 50t + 30^\circ)$$

$$u_2 = \sqrt{2} \cdot 2 \text{ V} \cdot \cos(2\pi 50t - 45^\circ)$$

$$u_3 = \sqrt{2} \cdot 5,563 \cos(2\pi 50t + 9,68^\circ)$$

KAPAZITÄT / KONDENSATOR

$\downarrow i$

KONDENSATOR / KAPAZITÄT

Speicherung von Ladung

$$C = \frac{Q}{U} \quad [C] = 1 \text{ F}$$

$$dQ = C \cdot dU$$

$$i = \frac{dQ}{dt} = C \cdot \frac{dU}{dt}$$

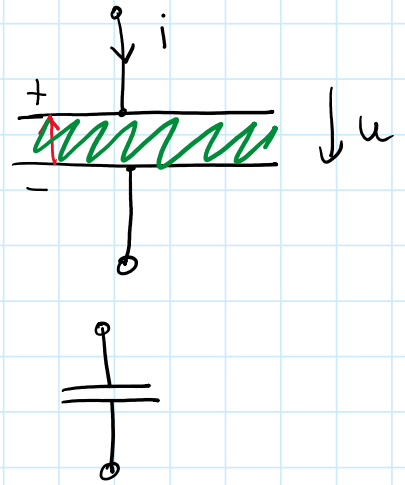
} isoliert

C hängt ab

- Plattenfläche
- Abstand der Platten
- Dielektrikum

Parallelschaltung $C_{\text{ges}} = \sum_{i=1}^n C_i$

Reihenschaltung $\frac{1}{C_{\text{ges}}} = \sum_{i=1}^n \frac{1}{C_i}$



SPULE / INDUKTIVITÄT L

$$[L] = 1 \text{ H}$$

$$U = L \cdot \frac{di}{dt}$$

